ALICE L WALTON SCHOOL OF MEDICINE

Primary (Essential) Hypertension

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Adapted from: Primary (Essential) Hypertension

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0:00 / 31:09 1x

Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of primary hypertension.
2. Explain pathophysiologic mechanisms that may underlie primary hypertension.
3. Justify the use of antihypertensive agents based on their mechanism of action.
4. Recognize common adverse drug reactions for antihypertensive agents and discuss ways to mitigate them.
5. Design an appropriate evidence-based treatment plan for hypertension treatment.

CASE CONNECTION

PM, a 58-year-old female, comes to the office to establish care. "I haven't been to a doctor in years," she says. "But I earn points at work for wellness visits." PM has no specific concerns. She smokes a pack of cigarettes a day, drinks two glasses of wine daily, and rarely exercises. PM's blood pressure is 146/80 mm Hg, and a repeat measure is 148/90 mm Hg. Her BMI is 28. Her physical exam is otherwise unremarkable.

How will you initially manage PM's elevated blood pressure? What can you tell her about long-term complications of uncontrolled blood pressure? Consider your answers as you read, and we'll revisit PM at the end of the brick.

GO TO CONCLUSION ↓

What Is Primary Hypertension?

High blood pressure, or hypertension, is the most common disease seen in US medical practice and the most common cardiovascular disease worldwide. Primary, or essential, hypertension is defined as **hypertension without an identifiable cause.** This contrasts with **secondary hypertension** in which a specific cause like hi is identified.

Primary hypertension represents 90% to 95% of all cases of persistently elevated blood pressure (Figure 1).

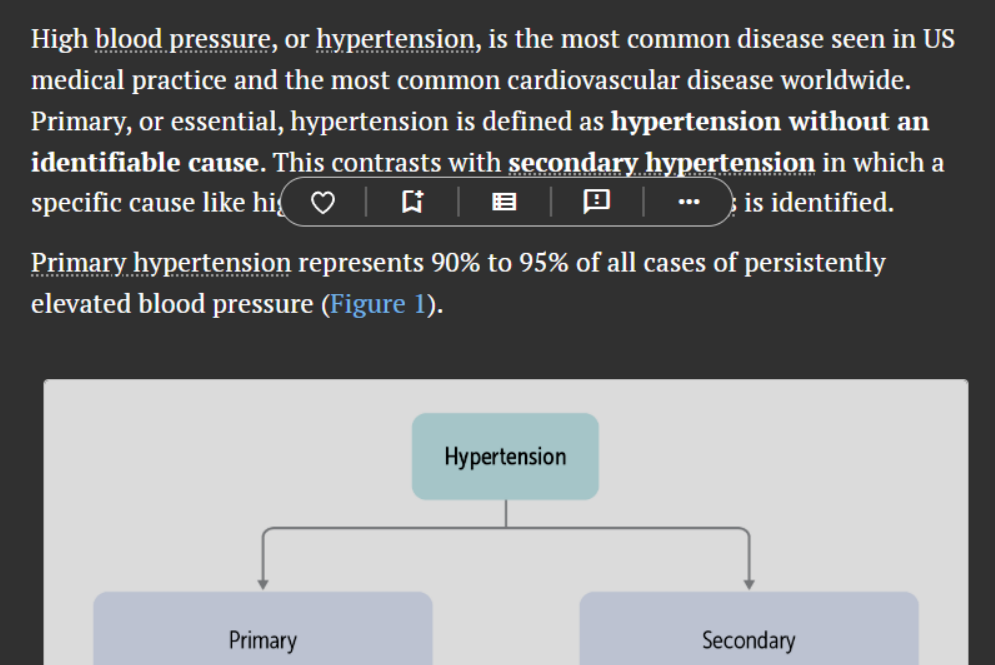
Hypertension
Primary
Secondary
90%-95% of cases
• Cause is unknown
• Results from complex interaction between genes, lifestyle, and environmental factors
5%-10% of cases
• Results from an identifiable cause: eg, kidney, renovascular, or endocrine disease; pregnancy, obesity, sleep apnea

Figure 1. Primary versus Secondary Hypertension.

What is the difference between primary hypertension and secondary hypertension?

How Do Patients with Hypertension Present? (LO #1)

Hypertension is nicknamed the "silent killer" because most patients are asymptomatic. It is usually discovered on routine checks of vital signs. For most patients, hypertension is defined as repeated blood pressure measurements of more than 130/80 mm Hg. The common lack of symptoms means that patients can go for years without knowing they are hypertensive or until cumulative organ damage occurs.

What is the most common presentation of hypertension?

What Is the Pathophysiology of Primary Hypertension? (LO #2)

The pathophysiologic mechanisms underlying primary hypertension (also called essential hypertension) are not fully understood and may vary from patient to patient, since this is not a discrete single disease but instead a clinical syndrome. Risk factors include both modifiable and nonmodifiable elements.

Etiology and Mechanism

Although no specific gene has been primarily linked to primary hypertension, a significant genetic component is likely, as primary hypertension is heritable.

Increased Systemic Vascular Resistance. Elevated systemic vascular resistance (SVR) is the most common mechanism for primary hypertension. This elevation has different causes in each patient, which explains why patients respond differently to antihypertensive medications.

Blood pressure is controlled either by cardiac output or SVR. The SVR can be affected by remodeling processes or thickening of the vessel walls, leading to stiffer walls (decreased compliance to stretch), or increased sympathetic tone and vasoconstriction. A narrower lumen is more resistant to blood flow, increasing the SVR. This process is most marked at the level of the **arteriole**. Many biologic, environmental, and genetic factors can increase SVR, but the overall mechanism is ause vasoconstriction versus those that cause vasodilation, ultimately leading to more vasoconstriction and higher SVR.

Catecholamines. One of the factors that can affect blood pressure, both directly and indirectly, is the sympathetic nervous system (SNS). The SNS and adrenal medulla release catecholamines such as norepinephrine and adrenaline, which bind to α-receptors on vascular smooth muscle. α1 or α2 -receptor activation causes vasoconstriction, which increases SVR and directly elevates blood pressure. It is thought that some patients either release too much or are simply too sensitive to catecholamines, ultimately resulting in hypertension.

Renin-aldosterone-angiotensin system. The sympathetic nervous system (SNS) also indirectly elevates blood pressure by activating the renin-aldosterone-angiotensin system (RAAS). The SNS causes the kidney to release renin, which increases angiotensin II (AT-II) and aldosterone levels. AT-II increases SVR through vasoconstriction of vascular smooth muscle. Aldosterone increases renal retention of sodium and water, leading to an increased blood volume. The increased volume leads to increased venous return to the heart and stroke volume, which means even more pressure within the vessels (Figure 2).

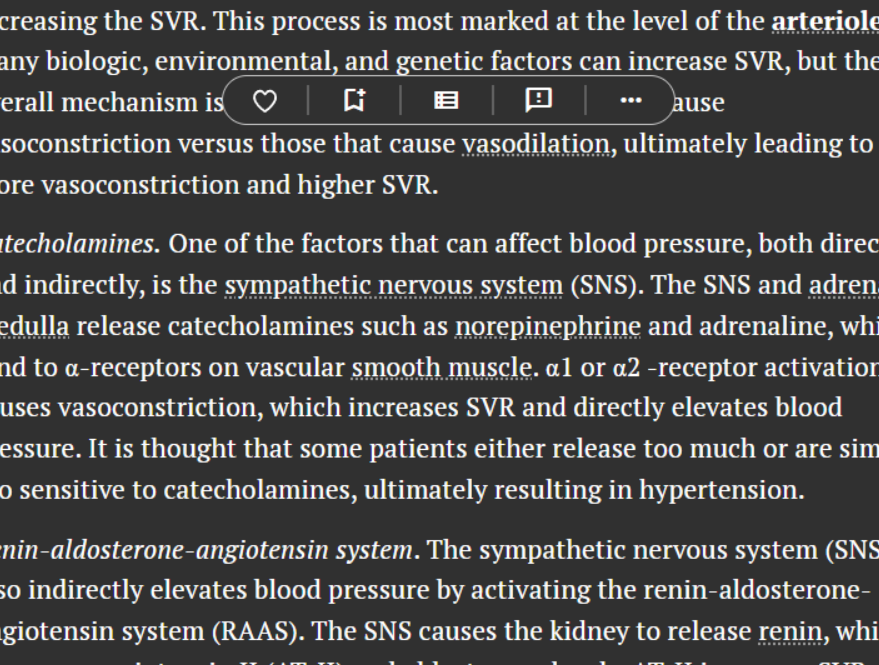

Figure 2. The Renin-Aldosterone-Angiotensin (RAAS) System. Note the AT-II increases SVR through vasoconstriction of vascular smooth muscle, whereas aldosterone increases blood pressure by causing renal reabsorption of sodium and water.

How do catecholamines increase blood pressure?

Decreased vascular compliance. With increasing age, small arteries harden and thicken, ultimately b en volume of blood entering the arteriole, the vessels can expand less, generate more resistance to flow, and the pressures increase. The loss of compliance increases SVR, and especially elevates the systolic blood pressure.

Increased Cardiac Output. Increased cardiac output due to increased stroke volume is a common cause of hypertension. This is usually due to **expanded blood volume** and occurs in a group of patients who are **"salt sensitive."** This means that increased salt intake leads to an exaggerated increase in their blood pressure. Why is this? In this group of patients, sodium retention expands extracellular fluid and blood volume. Increased blood volume increases stroke volume. In response to increased cardiac output, peripheral vessels constrict, raising SVR and contributing to elevated blood pressure. Why some patients are salt sensitive, and not others, is not well understood.

Increased cardiac output from an elevated heart rate is not the usual cause of primary hypertension; however, increased heart rate can transiently lead to elevated blood pressure. This most commonly occurs with exercise, caffeine consumption, and use of some drugs that elevate the heart rate (eg, cocaine, amphetamines).

The key factors that play a role in primary hypertension are summarized in Figure 3.

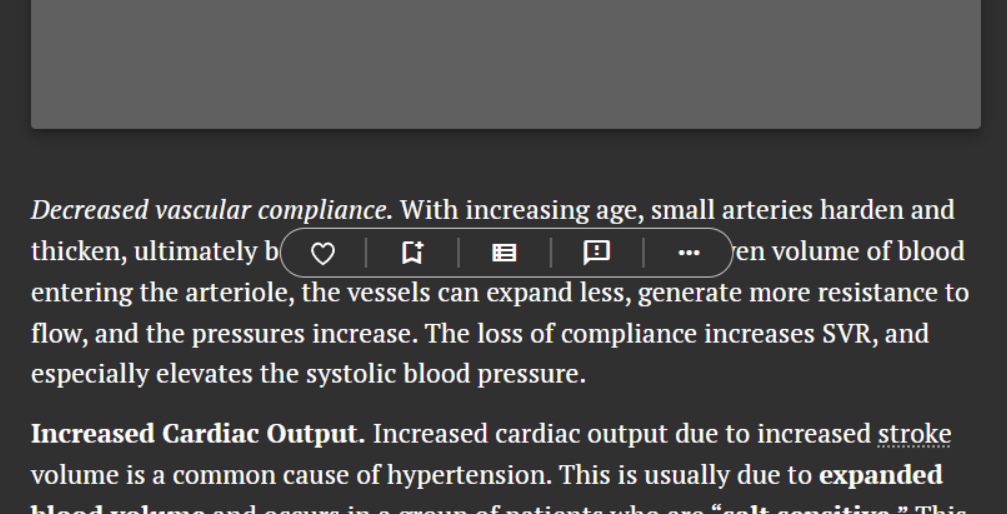

Figure 3. Common Factors that Influence Blood Pressure.

CLINICAL CORRELATION

Urinalysis is done in hypertensive patients to check for evidence of renal injury such as proteinuria and hematuria. Remember that uncontrolled hypertension can lead to narrowing of the small renal arteries which worsens the kidney's filtering function.

How Is Primary Hypertension Managed? (LO #4 and 5)

Treatment for hypertension includes a combination approach aimed at lifestyle modifications and, if necessary, pharmacologic intervention.

Lifestyle Changes

Lifestyle modifications remain the foundation of hypertension management across all risk catego ize that these interventions should be implemented concurrently with pharmacotherapy rather than sequentially. Key lifestyle recommendations include:

- Weight management: Aim for BMI <25 kg/m² through caloric restriction and increased physical activity
- Reduce intake of sodium and use of salt substitutes
- Alcohol cessation (in heavy drinkers)
- Enhance intake of Potassium-based salt substitutes
- Heart Healthy diets (Dietary Approaches to Stop Hypertension (DASH) diet): Rich in fruits, vegetables, whole grains, and low-fat dairy products
- Increase physical activity with a structured exercise program, including aerobic exercise and/or resistance training

What dietary modification can you make to help lower blood pressure?

How is the BP treatment determined? (LO#5)

The 2025 AHA guidelines emphasize that the selection of antihypertensive medications should be guided by a patient's overall cardiovascular disease (CVD) risk profile, comorbidities, and demographic factors. This personalized approach recognizes that certain medication classes may offer advantages beyond blood pressure reduction for specific patient populations. By matching pharmacological agents to individual risk profiles, clinicians can potentially improve outcomes and reduce adverse events.

The current guidelines combine the use of CVD risk estimation using the ASCVD-PREVENT risk calculator with BP levels to improve the benefits and effectiveness of anti-hypertensive therapy. The current guidelines recommend the treatment algorithm shown in Figure 4.

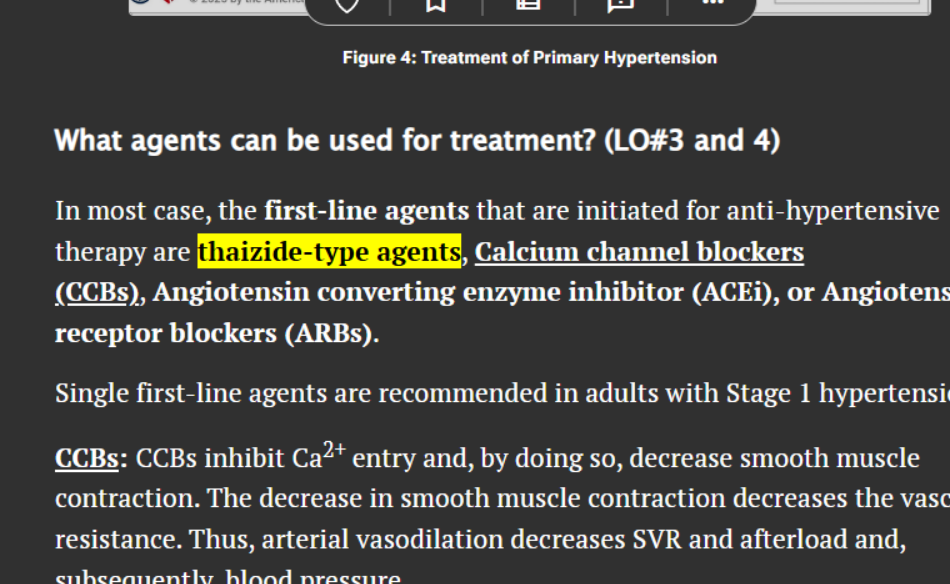
Use of Risk-Based Thresholds for Initiation of BP Treatment in Adults
BP Level-Only
Does the patient have an average BP ≥140/90 mm Hg?
YES
Initiate antihypertensive medications to lower BP and reduce CVD risk for primary or secondary prevention of CVD
NO
Risk-Based Thresholds for Initiation of BP Treatment for Adults*
Does the patient have existing clinical CVD (CHD, stroke, HF)?
YES
Initiate antihypertensive medications to lower BP and reduce CVD risk if average SBP ≥130 mm Hg or DBP ≥80 mm Hg for secondary prevention of CVD
NO
Does the patient have diabetes or CKD, or is the patient at increased short-term risk of CVD (10-year PREVENT-CVD risk ≥7.5%)?
YES
Initiate antihypertensive medications to lower BP and reduce CVD risk if average SBP ≥130 mm Hg or DBP ≥80 mm Hg for primary prevention of CVD
NO
Initiate antihypertensive medications to lower BP if average SBP ≥130 mm Hg or DBP ≥80 mm Hg after 3-6 months of lifestyle intervention
LEGEND
2025 High Blood Pressure
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Figure 4: Treatment of Primary Hypertension

What agents can be used for treatment? (LO#3 and 4)

In most case, the **first-line agents** that are initiated for anti-hypertensive therapy are **thaizide-type agents**, **Calcium channel blockers (CCBs), Angiotensin converting enzyme inhibitor (ACEi), or Angiotensin receptor blockers (ARBs).**

Single first-line agents are recommended in adults with Stage 1 hypertension.

CCBs: CCBs inhibit Ca^{2+} entry and, by doing so, decrease smooth muscle contraction. The decrease in smooth muscle contraction decreases the vascular resistance. Thus, arterial vasodilation decreases SVR and afterload and, subsequently, blood pressure.

CCBs are classified into dihydropyridines (DHP's) (E.g Nifedipine, Amlodipine) and non-dihydropyridine's (non-DHP's Verapamil and Diltiazem). Recall that DHPs are more potent at decreasing SVR, particularly long-acting DHPs such as amlodipine and felodipine. DHPs are used more often used in the treatment of hypertension. (review of CCBs from Foundations of Pharmacology here)

CLINICAL CORRELATION

DHPs such as Nifedipine are potent coronary vasodilators. They are used in the treatment of angina.

What are some potential adverse effects of CCBs? **Peripheral edema** from fluid extravasation secondary to increased capillary hydrostatic pressure in dilated vessels. **Reflex tachycardia (dihydropyridines only),** which occurs to compensate for the drop in blood pressure after rapid vasodilation, making dihydropyridines risky in most forms of ischemic heart disease. Constipation (with verapamil), lightheadedness, headache, and flushing can occur due to vasodilation.

Thiazide type diuretics: Diuretics are a class of medication that primarily increases diuresis, i.e. increase the excretion of water by the kidneys. The increased excretion of water decreases the volume, thereby decreasing blood pressure. Thiazides inhibit the Na-Cl symporter in the distal convoluted tubule, resulting in increased excretion of Na, K, Cl, and water by the kidneys, decreasing systemic volume and, subsequently, blood pressure. We will discuss the detailed mechanism of action of thiazides during the Renal course.

Examples of thiazide type diuretics include chlorthalidone, hydrochlorothiazide, chlorthalidone, indapamide, and metolazone.

What are the adverse effects from Thiazide-type diuretics use? Electrolyte disturbances (hypokalemia, hyponatremia), hyperuricemia, skin photosensitivity, rash.

ACE-inhibitors (ACEi) and ARBs: The inhibition of the ACE enzyme prevents the conversion of Angiotensin I to Angiotensin II. Angiotensin II plays a major role in the short-term and long-term regulation of blood pressure (Figure 2). ARBs block Ang-II binding to the AT1-R (Angiotensin-1 receptor). Blocking of AT1-R lowers the SVR and mean BP (both systolic and diastolic) in a hypertensive state and increases natriuresis (excretion of sodium in the urine).

Examples of ARBs inc -sartan)

The ACEi's inhibit the Angiotensin converting enzyme (ACE) decreasing the production of Ang-II, thus resulting in arteriolar dilation and increase the compliance of large arteriolar vessels. The decreased production of Ang-II results in decreased SVR. (review of ACEi/ARBs here from Foundations of pharmacology here)

CLINICAL CORRELATION

Often, ACEi/ARBs are combined with CCBs and diuretics to increase the diuretic response.

ACEi and ARBs decrease vasodilation of both arteries and veins; thus, reducing peripheral vascular resistance (decreases afterload) and increasing venous capacitance (decreases preload). This results in a decrease in afterload and preload.

Examples of ACE-I include ramipril, lisinopril, enalapril (ending with -pril).

What are the adverse effects from ACEi/ARB use? A **dry cough** caused by the accumulation of bradykinin can occur. A decrease in dose or switching to a different medication or class is effective to stop the dry cough. **Hyperkalemia:** The inhibition of aldosterone can lead to an increase in serum K. Thus monitoring K levels is recommended for these agents. Interactions with other agents, such as K-sparing diuretics, NSAIDs, can occur. **Angioedema** (more likely with ARBs) can also occur.

Warnings: These agents should be used with caution during pregnancy. ACEi and ARBs are teratogenic.

CLINICAL CORRELATION

Given their ability to dilate the efferent arteriole, ACEi and ARBs are often treat used to treat hypertension in patients with proteinuria.

In case of Stage-2 hypertension 2 first-line agents from different class can be chosen. In fact, a single pill, fixed-dose combination is preferred over two different pills. This has been shown to improve adherence. In specific cases, use of alternative agents is also considered.

Alternative anti-hypertensive agents

α-Adrenergic blockers used in the treatment of hypertension include prazosin, doxazosin, and terazosin. These agents produce a competitive blockade of α1-adrenoceptors. Recall that α1-receptors are expressed predominately in the smooth muscle vasculature and inhibition of the α1-receptors leads to a decrease in vasoconstriction. They decrease peripheral vascular resistance and lower arterial blood pressure by causing relaxation of both arterial and venous smooth muscle. α-blockers are no longer used as first-line agents in the treatment of hypertension.

CLINICAL CORREL

Reflex tachycardia and ***postural hypotension*** often occur at the onset of treatment and with dose increases, requiring slow titration of the drug.

What are the adverse effects from use α1- antagonists? **Orthostatic hypotension (OH)** is a significant adverse effect of α1-blockers. OH increases the risk of falls, particularly in older adults. All non-selective α1-blockers carry this risk. The risk is higher with the first dose, rapid increase in dose, and other medications such as nitrates and PDE-5 inhibitors. Prazosin has a higher risk due to its rapid onset of action. Priapism and prolonged erections have also been reported (associated with non-selective α1-blockers).

Clonidine acts centrally as an α2 agonist to produce inhibition of sympathetic vasomotor centers, which decreases sympathetic outflow to the periphery. This results in a reduction of total peripheral resistance and a decrease in blood pressure.

α-2 agonists are considered a last-line agent for the treatment of hypertension.

What are the adverse effects from use of Clonidine? Sedation, **dry mouth**, and constipation.

CLINICAL CORRELATION

Clonidine is used primarily for the treatment of hypertension that has not responded adequately to treatment with two or more drugs.

β-blockers These drugs work by reducing heart rate (negative chronotropic effect) and contractility (negative inotropic effect) via $β_1$ receptor blockade, which then reduces cardiac output and blood pressure. Blocking $β_1$ receptors also reduces renin release, which contributes to their antihypertensive effect. $β_1$-selective (e.g Metoprolol), non-selective agents (e.g Propranolol), and mixed α1 and β-antagonist (Labetalol) can be used. β-blockers are **not recommended** for first-line use for hypertension.(review of β-blockers from Pharmacology here)

What are the adverse effects from use β-1 blockers? Fatigue, Dizziness, Bronchospasms (non-selective β blockers), masking hypoglycemic symptoms

CLINICAL CORRELATION

β-blockers could be used to manage hypertension in patients with coronary heart disease or heart failure.

Minoxidil Minoxidil is metabolized to minoxidil sulfate, the active metabolite. This metabolite opens potassium channels in vascular smooth muscle, increasing potassium permeability and stabilizing the membrane near its resting potential. This leads to repolarization of the cell, which helps decrease vasoconstriction.

What are the adverse effects from the use of minoxidil? Headache, sweating, and hypertrichosis. When ons, such as β-blockers it can cause tachycardia (vasodilation --> causes baroreceptor reflex--> tachycardia).

The 2025 AHA guidelines advocate for a dynamic approach to treatment adaptation based on ongoing risk assessment and treatment response. This approach recognizes that a patient's risk profile may change over time due to aging, development of new comorbidities, or improvements in modifiable risk factors. Key principles for adapting therapy include:

- Reassess blood pressure control and overall cardiovascular risk at each follow-up visit
- Adjust medication dosages or add agents if blood pressure remains above target despite adherence
- Consider de-intensification of therapy in low-risk patients who maintain blood pressure well below target for an extended period
- Modify treatment approach if new comorbidities develop or existing conditions progress
- Reevaluate the risk-benefit balance of current therapy as the patient ages or develops frailty

For patients who fail to achieve blood pressure targets despite multiple medications at optimal doses (resistant hypertension), the guidelines recommend a systematic approach:

- Confirm true resistance by ruling out white-coat hypertension, medication nonadherence, and suboptimal dosing
- Screen for secondary causes of hypertension (e.g., primary aldosteronism, renal artery stenosis)
- Optimize the current regimen, ensuring it includes a diuretic appropriate for the patient's r

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to PM, how do you initially manage her elevated blood pressure? What do you tell her about long-term complications of uncontrolled blood pressure?

After you discuss a trial of lifestyle modification with PM—weight loss, moderation in alcohol consumption, exercise, smoking cessation, and sodium restriction—you explain why blood pressure control is important. You explain that chronic hypertension can lead to heart disease, chronic kidney disease, and stroke. PM takes your suggestions seriously. You schedule her for an office visit in 6 weeks to assess her response to lifestyle modification. Her blood pressure at that visit is 132/76 mm Hg. You opt to continue lifestyle modification and schedule another visit in 3 months.

Summary

- Primary hypertension is defined as elevated blood pressure (**≥130 mm Hg)** without an identifiable cause and is the most common form of hypertension.
- Hypertension is often asymptomatic and typically discovered during routine checks of vital signs.
- The pathophysiology of primary hypertension involves increased systemic vascular resistance, catecholamine sensitivity, and the renin-aldosterone-angiotensin system.
- The current treatment guidelines combine CVD risk with BP measurements to determine therapy.
- If BP is **≥140 mm Hg** or DBP **≥90 mm Hg** initiate anti-hypertensive medication.
- If BP is **≥130 mm Hg** or DBP **≥80 mm Hg** initiate anti-hypertensive medication and calculate the ASCVD-PREVENT risk score.
- Lifestyle modifications, such as weight management, reduced sodium intake, and increased physical activity, are foundational in managing hypertension.
- First-line treatments for hypertension include thiazide diuretics, calcium channel blockers, ACE inhibitors, and ARBs.
- The choice of antihypertensive medication may depend on individual patient factors, including comorbid conditions and specific risk factors.

Review Questions

1. A previously healthy 71-year-old male is diagnosed with new-onset hypertension after two office visits with readings of 145/88 mm Hg. Which of the following is the best initial treatment for this patient?

A. ○ Angiotensin-receptor blocker
B. ○ Angiotensin-converting enzyme inhibitor
C. ○ Calcium-channel blocker
D. ● Lifestyle modification
E. ○ Thiazide diuretic

Hide Explanation

The correct answer is lifestyle modification (D). The initial treatment of mild-to-moderate hypertension is lifestyle modifications such as diet, exercise, weight loss, and alcohol cessation. An angiotensin-receptor blocker (A) or angiotensin-converting enzyme inhibitor (B) would be an excellent first choice in patients who are hypertensive with diabetes, kidney diseases, or heart failure. A calcium-channel blocker (C) or thiazide diuretic (E) would be an appropriate first pharmacologic treatment in otherwise healthy individuals who did not respond to lifestyle modifications.

2. Which of the following best describes essential hypertension?

A. ○ Low blood pressure due to volume depletion.
B. ○ An increase in systemic vascular resistance that only effects the renal vessels.
C. ○ A temporary spike in blood pressure due to stress or exertion.
D. ○ Hypertension associated with an identifiable underlying medical condition.
E. ● Hypertension without an identifiable cause.

Hide Explanation

The correct answer is hypertension without an identifiable cause. This is in contrast to secondary hypertension, which is hypertension associated with an identifiable medical condition (D). Hypertension is an increase in blood pressure, not low blood pressure (A). Hypertension can be caused by an increase in systemic vascular resistance, but is a systemic disease and not focused on a single organ system (B). A temporary spike in blood pressure due to stress or exertion (C) can raise blood pressure, but this would not meet the definition for hypertension.

3. What are the two main pathophysiologic mechanisms that cause high blood pressure?

A. ○ Decreased SVR and/or decreased cardiac output
B. ○ Decreased SVR and/or increased cardiac output
C. ● Increased SVR and/or increased cardiac output
D. ○ Increased SVR and/or decreased cardiac output

Hide Explanation

The correct answer is (C) - the two main pathophysiologic mechanisms that increase blood pressure are to either increase the SVR, and/or to increase the cardiac output. Answers (A), (B), or (D) all involve decreases to one or both of these parameters, which would have the opposite effect of decreasing blood pressure.

4. A 65-year-old woman comes to clinic for her yearly check-up. She has no complaints, and does not currently take any medications. During her vital sign assessment, you record a blood pressure of 145/95 mm Hg. This is the first time the patient has had an elevated blood pressure in your office. The patient asks, "Do I have high blood pressure?" What is the best response?

A. ○ Yes, you have stage I hypertension, and we need to discuss changes to your diet and exercise
B. ○ Yes, you have stage II hypertension, and we need to discuss high blood pressure medications
C. ○ No, your blood pressure is fine. We will check it again next year.
D. ● We need to wait 2 weeks and then check it again. If it is still high at that time, then we can discuss our options.

Hide Explanation

The correct answer is (D). The key to this question is that the clinical diagnosis of hypertension requires repeated measurements demonstrated elevated an blood pressure, and should not rely on a single (isolated) measurement. A diagnosis of hypertension can be established with an average of at least two elevated blood pressure readings taken at least 2 weeks apart. Answers (A) and (B) are incorrect because they rely on a single blood measure measurement to arrive at a diagnosis of hypertension (although the correct categorization would be stage II hypertension, which is a systolic reading greater than or equal to 140 mm Hg or a diastolic reading greater than or equal to 90 mm Hg). Answer (C) is wrong because the patient does have an elevated blood pressure measurement that merits more immediate follow-up.

5. A 56-year-old female diagnosed with Stage 1 hypertension is prescribed an antihypertensive. During her three month visit, she reports that she has swelling around her ankles which she never had before. What antihypertensive agent is most likely to cause the swelling?

A. ● Felodipine
B. ○ Clonidine
C. ○ Lisinopril
D. ○ Metoprolol

Hide Explanation

The correct answer is (A) Felodipine. Felodipine is a non-DHP CCB that can cause peripheral edema due to fluid extravasation secondary to increased capillary hydrostatic pressure in dilated vessels. Choice (B), is incorrect; Clonidine most likely causes dry mouth. Choice (C), lisinopril, is incorrect since lisinopril use is associated with dry cough as an adverse effect. Choice (D), metoprolol is incorrect. Metoprolol, a β1-selective antagonist, is associated with fatigue, dizziness, and decreased HR (bradycardia).

How would you rate this Brick?

References

- Abrahamowicz, A. A., Ebinger, J., Whelton, S. P., Mensah, Y. C., & Yang, E. (2023). Racial and ethnic disparities in hypertension: Barriers and opportunities to improve blood pressure control. *Current Cardiology Reports*, *25*(1), 17–27. https://doi.org/10.1007/s11886-022-01826-x
- Bailey MA, Dhaun N. Salt Sensitivity: Causes, Consequences, and Recent Advances. Hypertension. 2024 Mar;81(3):476-489. doi: 10.1161/HYPERTENSIONAHA.123.17959. Epub 2023 Sep 18. PMID: 37721034; PMCID: PMC10863664.
- Iqbal, A. M., & Jamal, S. F. (2023, July 20). *Essential hypertension*. National Library of Medicine; StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK539859/
- https://www.uptodate.com/contents/choice-of-drug-therapy-in-primary-essential-hypertension?search=essential%20hypertension&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2
- https://www.uptodate.com/contents/overview-of-hypertension-in-adults?search=essential%20hypertension&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- Kim HL. Arterial stiffness and hypertension. Clin Hypertens. 2023 Dec 1;29(1):31. doi: 10.1186/s40885-023-00258-1. PMID: 38037153; PMCID: PMC10691097.
- Kumar V, Abbas AK, Aster JC. Heart. In: Kumar V, Abbas AK, Aster JC, eds. *Robbins Basic Pathology*. 10th ed. Elsevier; 2018: 399-440.
- Sutters M. Approach to Hypertension. In: Papadakis MA, McPhee SJ, Rabow MW, McQuaid KR. eds. *Current Medical Diagnosis & Treatment 2023*. McGraw Hill; 2023: e1-15.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of primary hypertension.~~
02 ☑ ~~Explain pathophysiologic mechanisms that may underlie primary hypertension.~~
03 ☑ ~~Justify the use of antihypertensive agents based on their mechanism of action.~~
04 ☑ ~~Recognize common adverse drug reactions for antihypertensive agents and discuss ways to mitigate them.~~
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Objective Confidence: 5/5